Milwee Math Competition – March 5, 2021 – **Geometry**

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Choose the best answer for each question below. Input your answers in the Google Form provided. For all questions, choice "E. NOTA" means "none of these answers are correct."

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1. The complement of angle A is 34° more than one-third of the measurement of angle A. What is the value of twice the measurement of the complement of the angle A?

   A) 84 B) 108 C) 144 D) 152 E) NOTA

2. A triangle is drawn on a graph such that its vertices are located at (1,2), (5,5), and (9,2). What is the area of the triangle (in units squared)?

   A) 8 B) 12 C) 14 D) 20 E) NOTA

3. A square piece of paper is measured and found to have side lengths of $(3 + x)$, $(-2x + (8 * 3))$, $(5x - 5^2)$, and $(2x - \sqrt[3]{64})$. Solve for x.

   A) 3 B) 7 C) 15 D) 21 E) NOTA

4. A right triangle has one leg that is 5 in. long and a hypotenuse that is 13 in. long. What is the length, in feet, of the missing leg?

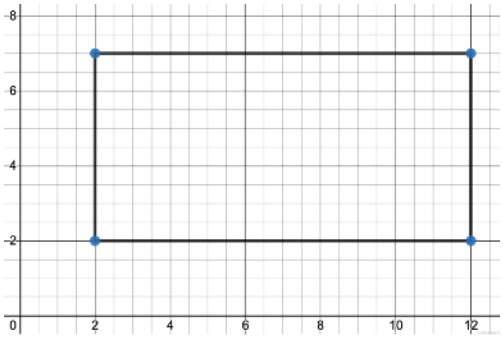
   A) 1 B) $\frac{13}{12}$ C) 12 D) 13 E) NOTA

5. At 5:48, how many degrees is the angle between minute and hour hand on an analog clock? *Note that the space between each minute is 6°*.

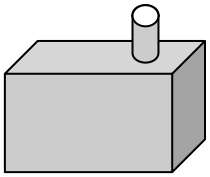
   A) 60° B) 96° C) 108° D) 114° E) NOTA

6. Which set of points identifies a rectangle that is similar to the one shown to the right?

   A) (1,3), (4,3), (1,2), (4,2)
   B) (12,5), (5,12), (12,12), (5,5)
   C) (5,10), (12,20), (5,20), (12,10)
   D) (6,1), (–2,5), (–2,1), (6,5)
   E) NOTA



7. A factory that produces boots for cats has a right cylindrical chimney, which allows smoke to escape the factory floor. What is the volume (in feet squared) of the chimney if its height is 15 feet and its diameter is 10 feet? Leave your answer in terms of π.



   A) 75π B) 150π C) 375π D) 750π E) NOTA

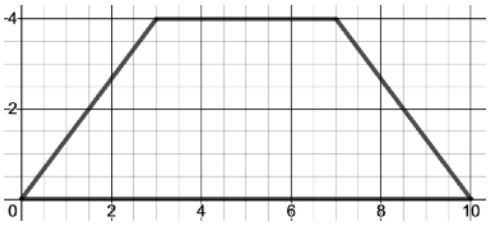
8. A triangular design on a carpet has two sides measuring 13 inches and 5 inches. Which of the following is a possible length of the other side?

   A) 4 in. B) 5 in. C) 6 in. D) 7 in. E) NOTA

9. If the measurements of three angles in a quadrilateral are 89°, 120°, and 75°, what is the measurement of the remaining angle?

   A) 64° B) 71° C) 76° D) 84° E) NOTA

10. An isosceles trapezoid has two bases measuring 10 feet and 4 feet and has a height of 4 feet. What is the length of one of the sides of the trapezoid?

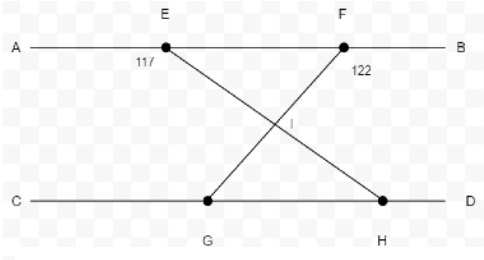


   A) 5 feet C) 6 feet E) NOTA
   B) $4\sqrt{2}$ feet D) 12 feet

11. A quadrilateral has angles A, B, C, and D. Angle A is equal to 67°, angle B is twice the complement of angle A, angle C is the supplement of angle B, and angle D is equal to the sum of angles A and B. What is the measurement of angle D?

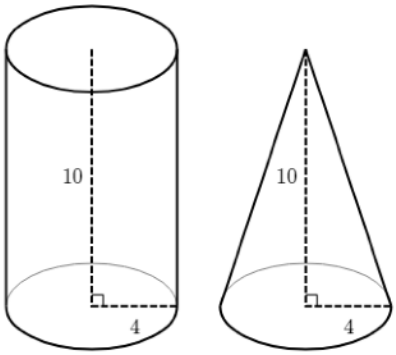
   A) 67° B) 90° C) 113° D) 157° E) NOTA

12. In the figure to the right, line AB is parallel to line CD, points E and F fall on line AB, and points G and H fall on line CD. What is the measurement of angle EIF?



   A) 58° C) 63° E) NOTA
   B) 59° D) 64°

13. Consider the following cylinder and cone. Marcel tried to use Cavalieri's principle to show that the two figures have the same volume.



"The base areas are the same. Therefore, corresponding cross-sections have the same area. So the volumes must be the same."

What is the first mistake Marcel made?

   A) The areas of both bases are not equal.
   B) It isn't true that corresponding cross-sections have the same area.
   C) Marcel did not establish that the heights are the same.
   D) The rule that states corresponding cross-sections having the same area means the volumes of their shapes are the same only works with the same type of shape.
   E) NOTA

14. Write the equation of a line that is parallel to $y = 6x + 1$ and that passes through the point (–7, 1).

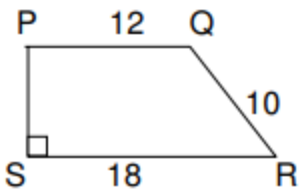
   A) $y = 6x + 43$ B) $y = -\frac{1}{6}x + \frac{1}{6}$ C) $y = \frac{1}{6}x + 43$ D) $y = 6x - 13$ E) NOTA

15. An ice cream company's top-selling flavor, "Chocolate Asteroid Crunch," is sold in a cylindrical container that is 16 centimeters tall and has a diameter of 14 centimeters. Based on suggestions from a focus group, the company decides to switch to a container that is 3.75 centimeters shorter so it fits better in consumers' freezers. If they want the total volume of ice cream in the new container to equal that of the old container, which of the following measurements (in centimeters) will best approximate the diameter of the new container?

    A) 15 B) 16 C) 17 D) 18 E) NOTA

16. Trapezoid PQRS has side lengths PQ=12, QR=10, and RS=18. m∠S = 90°. What is the area of PQRS (in units squared)?

    A) 120 B) 150 C) 180 D) 210 E) NOTA



17. Two coplanar circles, C_1 and C_2, share a common center. The radii of the circles are r and $r+2$ centimeters, respectively. The area outside of the smaller circle and inside of the larger circle is $32\pi^2$ centimeters. What is the sum of the circumferences of the two circles (in centimeters)?

    A) 32π B) 36π C) 40π D) 44π E) NOTA

18. Point P has coordinates (10, 0) on the xy-coordinate plane. Point Q is on the y -axis of the same plane. The slope of line PQ is −4. What is the distance between points P and Q?

    A) $10\sqrt{5}$ B) 30 C) $10\sqrt{17}$ D) 50 E) NOTA

19. One interior angle of a regular polygon with n sides has a measure of 175 degrees. What is the measure (in degrees) of the supplement of an interior angle of a regular polygon with $n/9$ sides?

    A) 30° B) 40° C) 42° D) 45° E) NOTA

20. A circle has a circumference of 18π cm and a center at point Q. An inscribed angle, ∠PQR, has a measure of 60 degrees. Which of the following is the length (in centimeters) of arc PQR?

    A) 3 B) 6 C) 3π D) 6π E) NOTA